

Chapter 11 Databases

Introduction

- Like a library, secondary storage is designed to store information
- End users need to understand
 - How information is organized in fields, records, tables and databases
 - The different types of databases and structures
- Competent end users need to be able to find information that is stored in databases

Data

Facts or observations about people, places, things, and events

- Used to be only numbers, letters, and symbols, but now also includes:
 - Audio, music, photographs, and video
- Two ways to view data
 - Physical view focuses on the actual format and location of data
 - Logical view focuses on the meaning, content, and context of the data

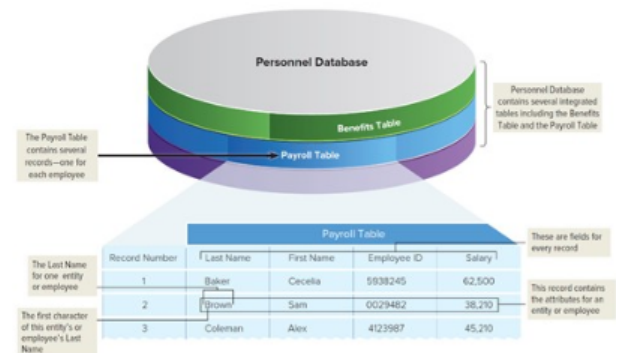
Data Organization

Logic view – data is organized into groups or categories

- Character – single letter, number, or special character
- Field – group of related characters
- Record – collection of related fields
- Table – collection of related records
- Database – integrated collection of logically related tables

Key Field or Primary Key

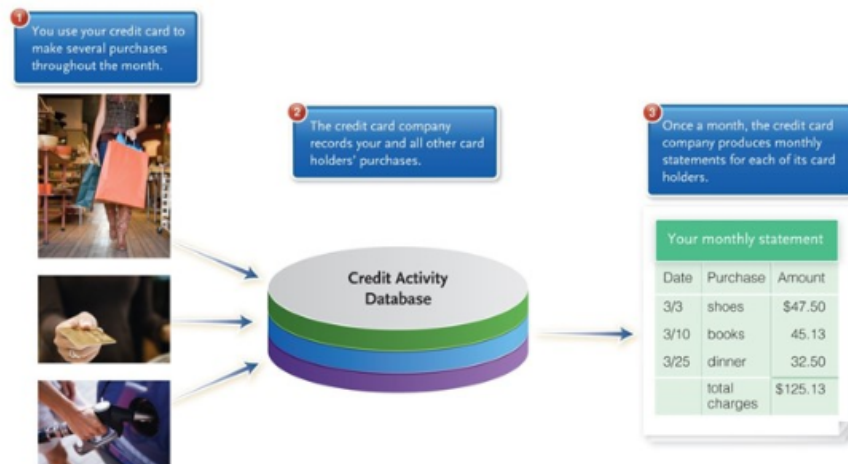
- Unique identifier
 - Used to create relationships between tables



Batch Processing

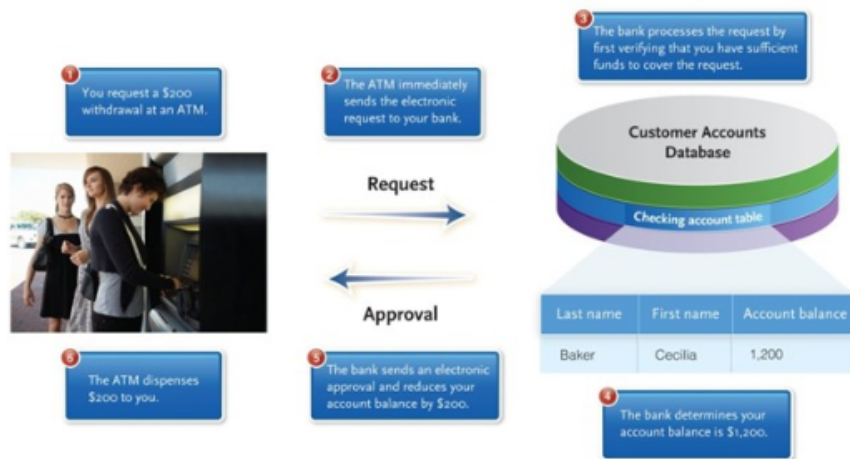
Data is processed in two ways: batch and real-time

- Data is collected over a period of time and the processing happens later all at one time



Real-Time Processing

Also known as online processing because it happens immediately during the transaction



Databases

- Collection of integrated data
 - Logically related files and records
- Databases address
 - Data redundancy – same information in multiple files
 - Data integrity – accurate updating of files

Advantages to having databases

- Sharing – between departments of an organization
- Security – limited access
- Less data redundancy – decrease unnecessary duplication
- Data integrity – reduce likelihood of inaccurate data

Database Management

Database Management System (DBMS)

- Software that enables users to create, modify, and gain access to data
- Software made up of:
 - DBMS engine – bridge between the logical view of data and the physical
 - Data definition subsystem – defines the logical structure by using:
 - Data dictionary or schema
 - Contains a description of the structure of data

The screenshot shows a form titled "Employee Records" with two main sections: "Personal Information" and "Contact Information". The "Personal Information" section includes fields for Employee ID (with a description: "Unique 3 digit number assigned to each employee"), Name (with a description: "Enter as month, day, year (for example, 1/1/2000)"), Address (with a description: "A 2 character abbreviation entered in capital letters, followed by a digit extension, if available (for example 01001 0000)"), City (with a description: "Enter as 3 digit ZIP code"), State (with a description: "Enter as 2 digit state code"), and Zip (with a description: "Enter as month, day, year (for example, 1/1/2000, 01/01/2000, or 1/1/2001)"). The "Contact Information" section includes fields for Phone (with a description: "Enter as (000) 000-0000") and Email (with a description: "Enter as month, day, year (for example, 1/1/2000, 01/01/2000, or 1/1/2001)").

DBMS (Cont.)

- Data manipulation subsystem provides tools for maintaining and analyzing data
 - Data Maintenance – maintaining data
 - Analysis Tools used to view parts of the data
 - Query-by-example (QBE)
 - Structured query language (SQL)
- Application generation subsystem provides tools to create data entry
- Data administration subsystem helps manage the overall database
 - Database Administrators (DBAs) administer the database
 - Processing rights to determine who has access to the databases

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DBMS Structure

DBMS Programs are designed to work with data that is logically structured or arranged in a particular way

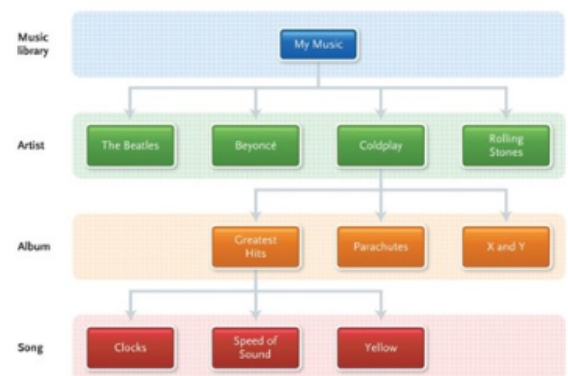
- Database model
 - Model defined rules and standards for data in a database

Organization	Description
Hierarchical	Data structured in nodes organized like an upside-down tree; each parent node can have only one parent
Network	Like hierarchical except that each child can have several parents
Relational	Data stored in tables consisting of rows and columns
Multidimensional	Data stored in data cubes with three or more dimensions
Object-oriented	Organizes data using classes, objects, attributes, and methods

Hierarchical Database

Fields or records structured in nodes

- Nodes - points connected like branches of an upside-down tree
 - One parent per node
 - Parent can have several child nodes
 - One-to-many relationship

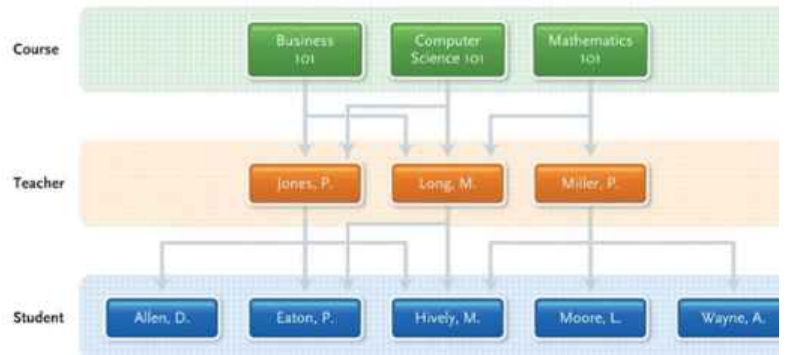


Major concern is that if your parent node is deleted then so are all subordinate child nodes

Network Database

Has a hierarchical node arrangement

- Each child node may have more than one parent node
 - many-to-many relationship
- Additional connections between parent and child are *Pointers*
- Nodes can be reached through multiple paths



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Relational Database

A more flexible type where there are no access paths down a hierarchy

- Data stored in table called a relation
- Tables linked via a common data item
- Tables consist of rows and columns

The diagram shows three tables: Vehicle Owner Table, Vehicle Table, and Outstanding Citation Table. The Vehicle Owner Table has columns: License Number, Last Name, First Name, Street, City, State, and Zip. The Vehicle Table has columns: License Number and Vehicle. The Outstanding Citation Table has columns: License Number and Citation Code. Arrows indicate relationships: License Number in Vehicle Owner Table is related to License Number in Vehicle Table; License Number in Vehicle Table is related to License Number in Outstanding Citation Table.

License Number	Last Name	First Name	Street	City	State	Zip
JF1987	Aaron	Linda	132 Park Lane	San Jose	CA	9512
D12372	Abar	John	139 Oak St.	Lakeville	CA	9412
LK3457	Abell	Jack	9587 State St.	Stone	CA	9320
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License Number	Vehicle
LK3457	1997 Ford Explorer
BD1396	2009 Toyota Camry
JR1321	2008 Chevy Suburban
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License Number	Citation Code
SK9847	Speed
DPL415	Park
LK3457	Speed
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Multidimensional Database

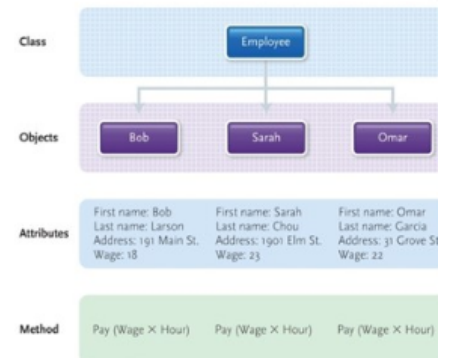
A variation and an extension of the relational model

- Data cube
 - Extension of the two dimensional data model to include additional or multiple dimensions
- Good for representing complex relationships
- Advantages over relational databases
 - Conceptualization provides users with an intuitive model in which complex data and relationships can be conceptualized
 - Processing speed for analyzing and querying a large multidimensional database is faster

Object-Oriented Database

Store data as well as instructions to manipulate data

- Organize data using
 - Classes – general definitions
 - Objects – specific instances of class containing data and instructions to manipulate the data
 - Attributes – data fields an object possesses
 - Methods – instructions for retrieving or manipulating attribute values



Types of Databases

Type	Description
Individual	Integrated files used by just one person
Company	Common operational or commonly used files shared in an organization
Distributed	Database spread geographically and accessed using database server
Commercial	Information utilities or data banks available to users on a wide range of topics

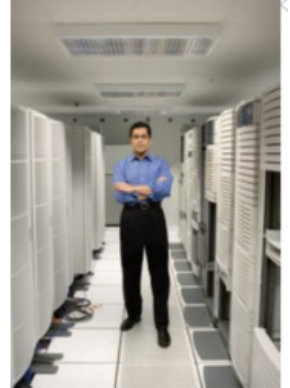
- Commercial databases
 - ProQuest Dialog
 - Dow Jones Factiva
 - LexisNexis

Database Uses and Issues

- Strategic uses
 - Data warehouse – storing in a database for special use
 - Data mining – used to search data warehouses
- Security
 - Databases are valuable so protection necessary
 - Protected by firewalls

Careers In IT

- Database administrators
 - Determine the most efficient ways to organize and access a company's data
 - Responsible for database security and backing up the system
- Employers look for
 - Bachelors degree in Computer Science or Information Systems
 - Technical experience
- Database administrators can expect to earn \$56,000 to \$90,000 annually



A Look to the Future ~ The Future of Crime Databases

- Have you ever imagined a world without violent crime?
 - Research institutions have been looking into patterns that could predict criminal behavior
 - However; not all states and governments are collecting the same data